End-to-End Computer Vision Pipeline for Perforation Defect Detection and Classification in Industrial Imaging

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Abstract

- Manual inspection of perforation regions is slow, subjective, and hard to scale.
- We build one end-to-end vision pipeline that localizes each region and assigns multiple defect labels per region.
- The result reaches 99.08% micro-F1: accurate, label-efficient, and fast for production-scale monitoring.

Problem Statement

Objectives

- Detect and localize perforation regions. (Fig1)
- Classify each region across multiple defect types.
- Reach production-scale accuracy and throughput.

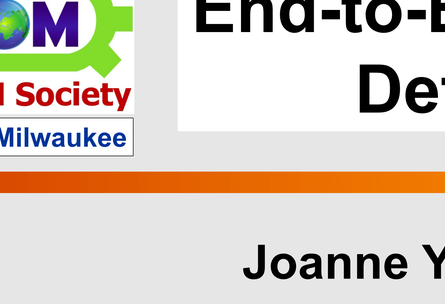


Fig 1. Four perforation classes
(a) DT0: non-defective (b) DT1: missing
(c) DT2: touching (d) DT3: out-of-bounds

Dataset

- Multiple defect types can co-occur within a region (DT2 & DT3)
- Small set of labeled images and a large pool of unlabeled images

→ Data curation

Labeling every image is costly, so we curate a compact training set from a large unlabeled pool, using active learning and bootstrapping to cover both common and rare defect cases.

Limitations

- No test-set ground truth (metric-based eval only)
- Production analysis restricted by partner confidentiality

Methodology

Image Selection (AL + Bootstrapping)

- Active-learning score: prioritize the sharpest, most edge-rich images for annotation[1,2].

$$s(I) = \mathrm{Var}(\nabla^2 I) = \frac{1}{N}\sum_{x,y}\left[\nabla^2 I(x,y) - \mu\right]^2$$

(I is the input image, ∇²I its Laplacian, μ the mean Laplacian, and N the number of pixels)

- Bootstrapping: adds a random subset of harder, low-information images (blur, lighting, deformation) to keep the model robust.

Iterative Refinement

- Round 1: train an initial EfficientNetB0 classifier on the small labeled set.
- Round 2: auto pseudo-label on selected images → manually verify → retrain on combined set (same auto-label-then-verify strategy as adaptive thresholding).

★ Adaptive Thresholding (Auto Pseudo-Labeling)

- Cuts annotation burden: auto-generates detection labels; annotators only fix wrong boxes.
- Dual-stage detector: Gaussian adaptive threshold + morphology, with Black-hat/Otsu fallback for low contrast & uneven lighting.
- Dot-association + NMS: captures out-of-bounds (DT3) defects, de-duplicates into clean labels.

End-to-End Pipeline

- Phase1: Detect and Crop / Phase 2: Multi-Label Classify & Aggregate

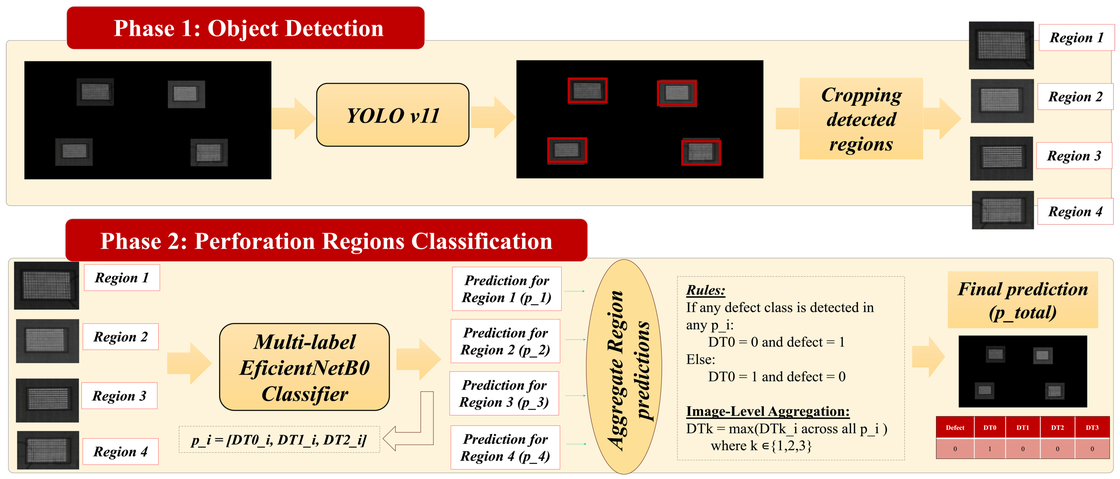


Fig 2. End-to-end detection-then-classification pipeline

Two-Model Design & Training

- ImageNet-pretrained EfficientNetB0[3], progressively fine-tuned (frozen to fully unfrozen), with augmentation and class-weighted binary cross-entropy
- YOLOv11[4] + EfficientNetB0 combines fast localization with fine-grained defect classification[5].

Results

- **Object Detection**

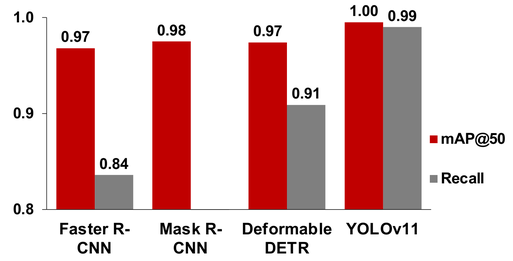


Fig 3. mAP@50 and Recall across Object Detectors[6,7,8]

- **Classification Accuracy**
  EfficientNetB0 multi-label, 99.08% micro-F1 / 98.37% macro-F1 across DT0-DT3
- **Auto-Annotation Efficiency**
  Adaptive thresholding correctly auto-labeled 93% of images
- **Efficiency & Rigor**
  8,844 images in 36 min; bootstrap 95% CI + McNemar confirm YOLOv11's gain

Conclusions

- End-to-end pipeline detects perforation regions (YOLOv11) and multi-label-classifies each region (EfficientNetB0).
- 99.08% micro-F1 with fast, large-scale inference, reached with far less labeling via AL + Bootstrapping
- **A real-time prototype** runs the pipeline on streaming frames, showing deployability for live line monitoring.
- Future work: production deployment with closed-loop control and more defect types.

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